

Likelihood and Bayesian Inference for the Efficacy of the COVID-19 Vaccine

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Abstract

The BNT162b2 mRNA vaccine developed by Pfizer and BioNTech received Emergency Use Authorization from the FDA in December 2020, following a clinical trial reported by Polack et al. (2020). In this paper, we analyzed the trial data using both likelihood (frequentist) and Bayesian frameworks to estimate vaccine efficacy ψ and check whether it exceeds the FDA's minimum threshold of 30%. Modeling the number of vaccine-group infections among the total 170 observed cases as a binomial random variable, we obtained a maximum likelihood estimate of $\hat{\psi}^{mle} = 95.1\%$ with a 95% confidence interval of (91.7%, 98.4%). Under a pessimistic Bayesian prior, the posterior median efficacy was 75.2% with a 95% highest posterior density interval of (66.9%, 82.2%). Both a likelihood ratio test and the posterior probability $P(\psi > 0.30 \mid t) = 1$ provided strong evidence that vaccine efficacy exceeds the FDA's threshold.

Keywords

Maximum Likelihood Estimation, Bayesian Inference, Covid-19, Vaccine Efficacy

Introduction / Background

In December 2020, the BNT162b2 vaccine made by Pfizer and BioNTech received an Emergency Use Authorization (EUA) from the FDA, which was a really big deal in the fight against the COVID-19 pandemic. This vaccine was a two-dose shot and it worked differently than some of the other vaccines being developed at the time. For example, vaccines from companies like Janssen and AstraZeneca used a viral vector approach, but BNT162b2 instead used something called mRNA, which is wrapped in lipid nanoparticles and works by teaching your immune system to recognize the spike protein on the virus. To figure out whether this new type of vaccine actually worked, a large clinical trial was run across multiple countries and published by Polack et al. (2020). The trial was placebo-controlled and double-blinded, meaning neither the participants nor the researchers knew who got the real vaccine. Over 34,000 people were randomly split into two equal groups, one got the vaccine and one got a placebo, in a 1:1 ratio. The study was set up to end once 170 total confirmed COVID-19 cases were recorded among people who had no prior signs of infection. While Pfizer and BioNTech specified a Bayesian approach in their trial protocol, using a Beta prior for the transformed efficacy parameter, the statistical merits and interpretation of this choice have been the subject of discussion in the literature (Senn, 2021). Notably, the trial was designed to stop once a fixed number of infection events were observed, a feature that motivates the conditional binomial model used in this analysis. When those 170 cases were counted up, only 8 of them came from the vaccine group ($n_1 = 17411$), while 162 came from the placebo group ($n_2 = 17511$). That works out to an infection rate of about 0.046% for the vaccinated group versus 0.925% for the placebo group, which is a striking difference.

	COVID-19 Cases	Healthy	Total
BNT162b2 (Vaccine)	8	17,403	17,411
Placebo	162	17,349	17,511
Total	170	34,752	34,922

Table 1: COVID-19 Infections by Treatment Group

Figure 1: Infection Rate by Group

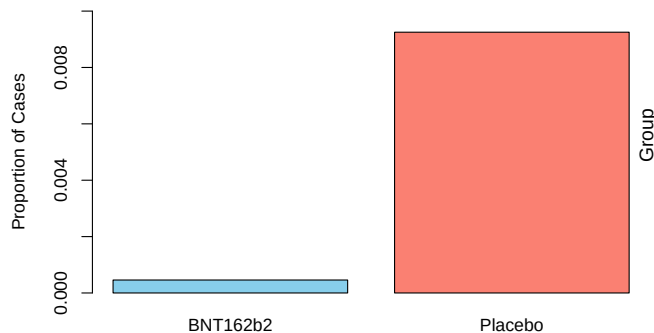
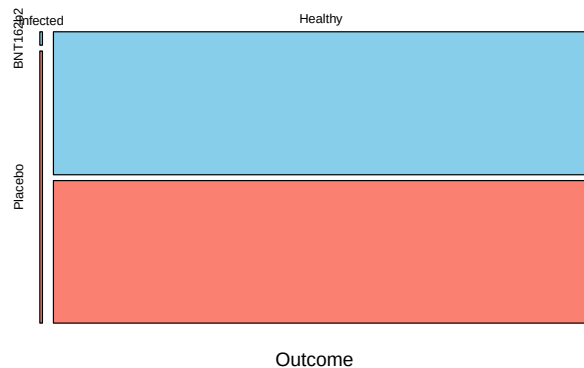


Figure 2: Mosaic Plot of Distribution



This analysis seeks to evaluate the efficacy of the vaccine based on these findings. Specifically, we test whether the vaccine efficacy (ψ) is significantly greater than the 30% threshold required for regulatory approval. We show our research goal through the hypotheses $H_0 : \psi = 0.30$ versus $H_1 : \psi \neq 0.30$.

Statistical Methods

Model

Let T denote the number of infections observed in the vaccine group among the total $n = 170$ infections recorded in the trial. We assume that T follows a binomial distribution (see Appendix for proof),

$$T \sim \text{Binom}(n = 170, \pi),$$

where $\pi = P(\text{vaccine} \mid \text{COVID-19 infection})$.

The observed data yielded $t = 8$ infections in the vaccine group and 162 infections in the placebo group.

The parameter of interest is the vaccine efficacy ψ , defined as

$$\psi = 1 - \frac{\pi_v}{\pi_p} = \frac{\pi_p - \pi_v}{\pi_p},$$

where π_v and π_p denote the probabilities of infection in the vaccine and placebo groups respectively.

In terms of the binomial probability π as follows

$$\psi = \frac{1 - 2\pi}{1 - \pi}$$

where $-\infty < \psi < 1$.

We are interested in an interval estimate for this parameter.

Also, we wish to test

$$H_0 : \psi = 0.3 \quad \text{versus} \quad H_1 : \psi \neq 0.3$$

as the FDA requires at least 30% efficacy for a new therapy to be approved.

Two statistical frameworks, Likelihood (frequentist) and Bayesian approach, are used to make inference on ψ as below.

Likelihood Inference

For the likelihood approach, we work directly with the vaccine efficacy parameter ψ rather than π . Since standard errors and significance tests are not invariant under transformation, we reparameterize the likelihood in terms of ψ by substituting $\pi = \frac{1-\psi}{2-\psi}$, and $1 - \pi = \frac{1}{2-\psi}$. This gives the reparameterized likelihood function

$$L^*(\psi) = \binom{n}{t} \left(\frac{1-\psi}{2-\psi} \right)^t \left(\frac{1}{2-\psi} \right)^{n-t},$$

with log-likelihood

$$\ell^*(\psi) = t \log(1 - \psi) - n \log(2 - \psi).$$

The MLE of ψ is found by differentiating $\ell^*(\psi)$ and setting the derivative equal to zero, yielding the closed form

$$\hat{\psi}^{MLE} = \frac{n - 2t}{n - t}.$$

This is a maximum is confirmed by the second derivative test

$$\frac{d^2 \ell^*}{d\psi^2} = -\frac{t}{(1 - \psi)^2} - \frac{n}{(2 - \psi)^2},$$

which is strictly negative for all $t > 0$, $n > 0$, and $\psi \in (-\infty, 1)$.

To construct a large sample confidence interval for ψ , we use the Fisher information.

By Theorem 12.1, for large n , the MLE is approximately normally distributed with standard error $1/\sqrt{I(\hat{\psi})}$, giving an approximate 95% confidence interval for ψ :

$$\hat{\psi}^{mle} \pm z_{0.025} \cdot \frac{1}{\sqrt{I(\hat{\psi}^{mle})}}.$$

This approximation is verified by comparing $\ell^*(\psi)$ against its second order Taylor approximation. Since ψ is a transformation of π , the quality of the normal approximation can be also assessed on the π scale, using the original log-likelihood $\ell(\pi)$, where the quadratic approximation is easier to observe.

Both comparisons are shown in figure below. The log-likelihood $\ell(\pi)$ and its Taylor approximation agree closely near $\hat{\pi} = t/n$, confirming that the normal approximation is justified for these data.

Figure3: Second Order Taylor Approximation

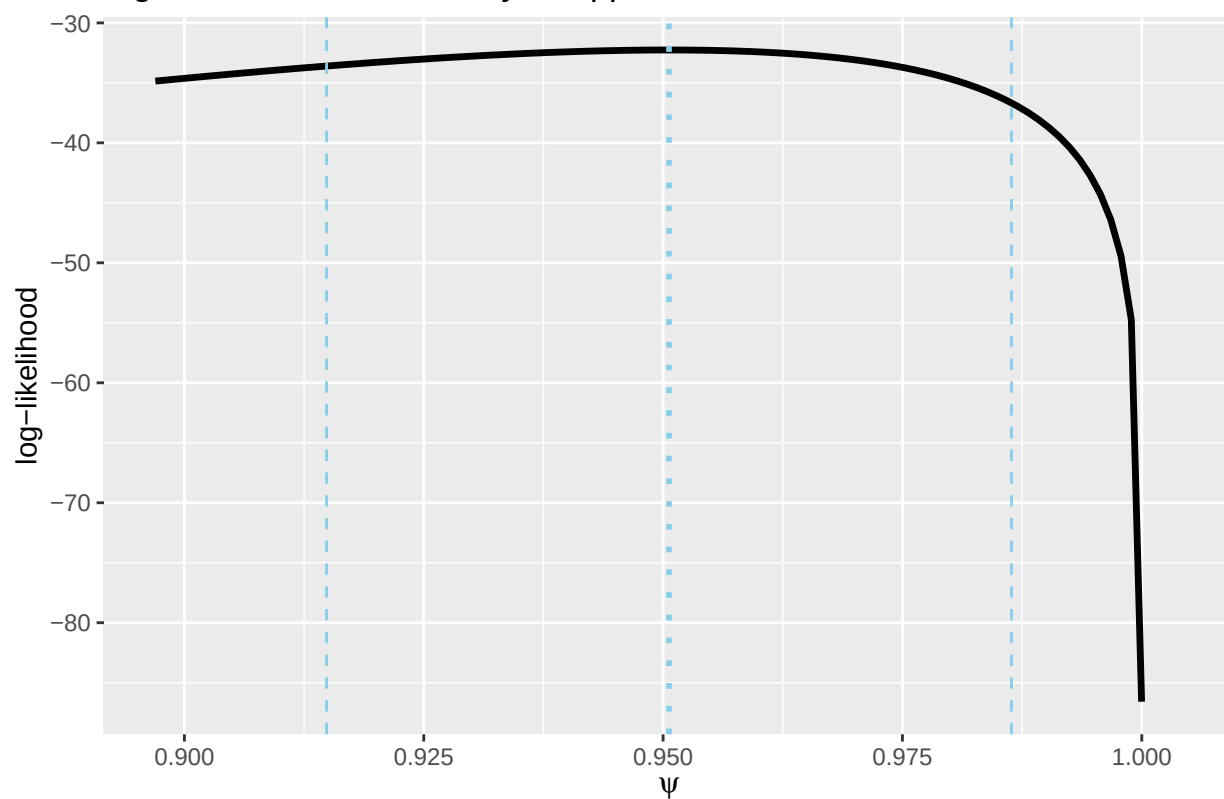
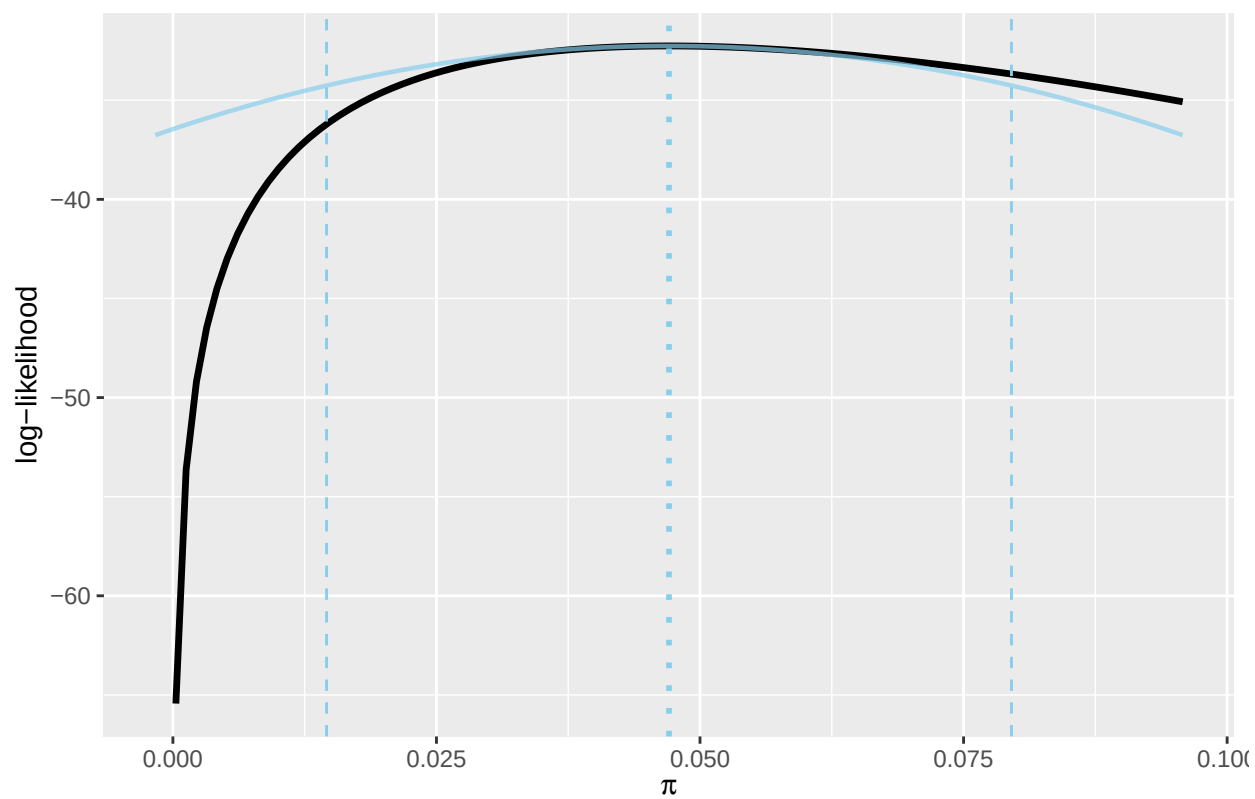


Figure4: Second Order Taylor Approximation



The parametric bootstrap percentile interval is also computed. Bootstrap samples $T^* \sim \text{Binomial}(n, \hat{\pi})$ are generated using $\hat{\pi} = t/n$, and the bootstrap estimator

$$\hat{\psi}^* = \frac{n - 2T^*}{n - T^*}$$

is calculated for each of $B = 10000$ replicates.

To assess whether the vaccine meets the FDA regulatory threshold, we test

$$H_0 : \psi = 0.30 \quad \text{versus} \quad H_1 : \psi \neq 0.30.$$

The likelihood ratio test statistic is $W = 2[\ell^*(\hat{\psi}^{mle}) - \ell^*(\psi_0^{null})]$, where the null value $\psi_0 = 0.30$ corresponds to $\pi_0 = \frac{1-\psi_0}{2-\psi_0}$. Under H_0 , the p-value is the right tail probability from the χ_1^2 .

An empirical p -value is also computed by simulating $B = 10000$ datasets under H_0 using $T^* \sim \text{Binomial}(n, \pi_0)$, recalculating W^* for each replicate, and taking the proportion of replicates for which $W^* \geq w$.

Bayesian Inference

For the Bayesian approach, we first need to choose a prior distribution for π , which basically just means we need to say what we believe about π before we even look at the data. We chose to use a Beta prior, so $\pi \sim \text{Beta}(a, b)$, which is a convenient choice because when you use it with a binomial likelihood, the posterior ends up being a Beta distribution too. To actually figure out what values to use for a and b , we did something called prior elicitation, where we started from a “pessimistic” view of how well the vaccine works. We set the prior median of π to 0.5, which means we’re starting with the assumption that a vaccinated and unvaccinated person are equally likely to get infected. We also said there’s only a 5% chance the vaccine meets the FDA’s required 30% efficacy threshold, so $P(\psi \geq 0.30) = 0.05$. To convert that into something usable on the π scale, we used the transformation $\pi = \frac{1-\psi}{2-\psi}$, which gives us a 5th percentile of about 0.4118. Plugging

both of those constraints in gave us:

$$a = 43.04, \quad b = 43.04$$

The likelihood function for π comes from the binomial distribution:

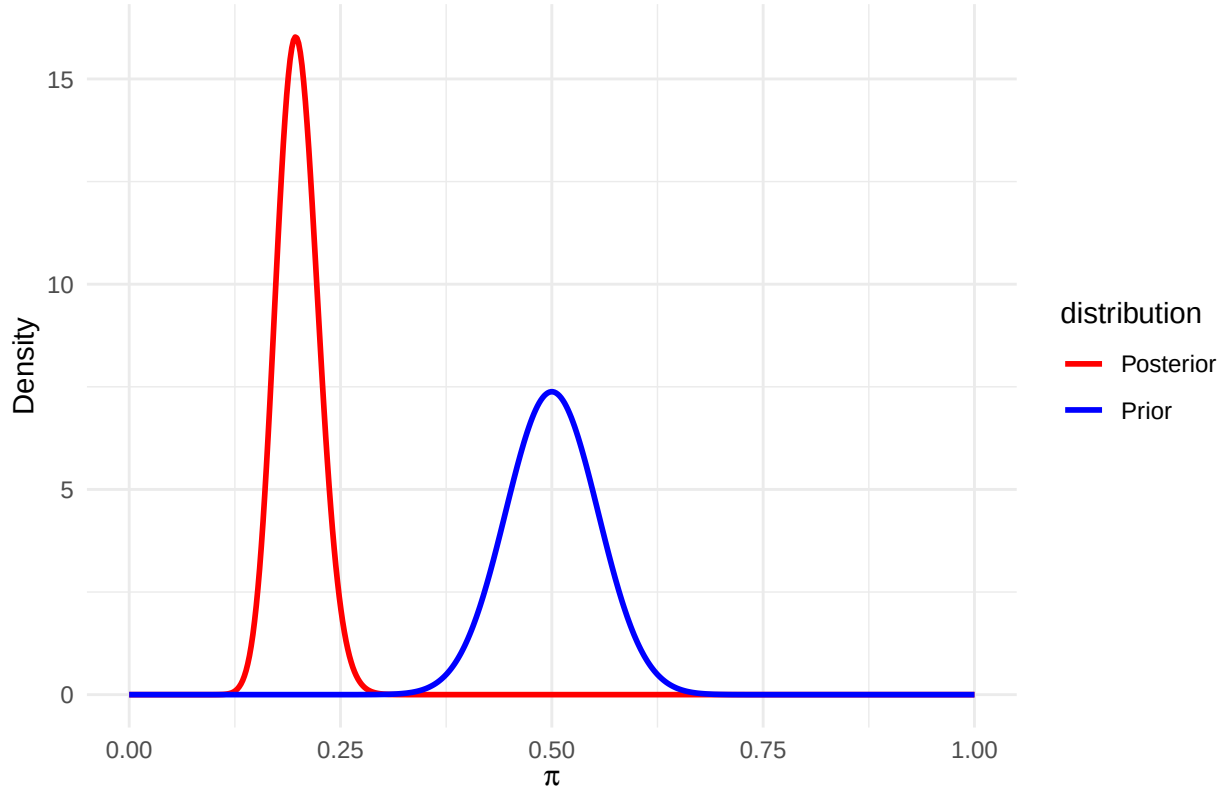
$$f(t \mid \pi) = \binom{n}{t} \pi^t (1 - \pi)^{n-t}$$

By Bayes' Theorem, the posterior is proportional to the likelihood times the prior. Due to the conjugacy of the Beta-binomial model, the posterior is also a Beta distribution with updated parameters:

$$\pi \mid t \sim \text{Beta}(a + t, b + n - t)$$

From the posterior distribution of π , we obtain the posterior median, a 95% Highest Posterior Density Interval (HPDI), a 95% equal-tail credible interval, and the posterior probability $P(\psi > 0.30 \mid t)$, all transformed back to the ψ scale using $\psi = \frac{1-2\pi}{1-\pi}$.

Figure5: Prior and Posterior Distribution



Results

Under the likelihood framework, the MLE of vaccine efficacy was $\hat{\psi}_{MLE} = \frac{154}{162} \approx 0.951$, corresponding to an estimated efficacy of 95.1%. The Fisher information at $\hat{\psi}$ was $I(\hat{\psi}) = 3434.88$, yielding a standard error of 0.0171. The large sample 95% confidence interval for ψ , constructed directly on the ψ scale using the Fisher information, was (0.917, 0.984). The parametric bootstrap percentile interval, based on $B = 10000$ replicates, was (0.910, 0.982). The two intervals are compared alongside the Bayesian credible intervals in Table 2.

For the likelihood ratio test of $H_0 : \psi = 0.30$ versus $H_1 : \psi \neq 0.30$, the observed test statistic was $W = 121.60$. The asymptotic p -value from the χ^2_1 distribution was $p = 2.82 \times 10^{-28}$. The empirical p -value, obtained by simulating $B = 10000$ datasets under H_0 , was $p = 0$, meaning none of the simulated statistics under H_0 equaled or exceeded the observed value of $W = 121.60$.

Under the Bayesian framework, the posterior distribution of π given the observed data is $\pi | t \sim \text{Beta}(51.04, 205.04)$, obtained by updating the pessimistic prior $\text{Beta}(43.04, 43.04)$ with $t = 8$ vaccine infections out of $n = 170$ total cases. The posterior median of π was 0.199, corresponding to a posterior median vaccine efficacy of $\hat{\psi} = 75.2\%$ on the ψ scale. The posterior probability $P(\psi > 0.30 | t)$ was found to be 1, indicating that the entire posterior distribution lies above the FDA efficacy threshold of 30%. The 95% equal-tail credible interval and 95% HPDI for ψ are reported alongside the frequentist intervals in Table 2.

Frequentist		Bayesian	
Large Sample CI	Bootstrap CI	Equal-Tail CI	HPDI
(0.917, 0.984)	(0.910, 0.982)	(0.666, 0.820)	(0.669, 0.822)

Table 2: Frequentist and Bayesian 95% interval estimates for ψ

Discussion / Conclusion

Both the likelihood and Bayesian analyses provide strong evidence that the BNT162b2 vaccine is highly effective against COVID-19, though the two approaches tell somewhat different stories about just how effective it is. The likelihood approach estimated vaccine efficacy at 95.1%, with both the large sample confidence interval (91.7%, 98.4%) and the bootstrap interval (91.0%, 98.2%). The

likelihood ratio test produced an essentially zero p -value under both the asymptotic and empirical approaches, providing evidence against the null hypothesis $H_0 : \psi = 0.30$. This is consistent with the 95% efficacy reported by Polack et al. (2020). The Bayesian approach told a more conservative story, with a posterior median efficacy of 75.2% and a 95% HPDI of (66.9%, 82.2%). This is lower than the MLE because of the pessimistic prior $\text{Beta}(43.04, 43.04)$. This was intentional. By starting with a skeptical prior and still finding strong evidence for the vaccine, the conclusion becomes more concrete. The posterior probability $P(\psi > 0.30 \mid t) = 1$ confirms that the entire posterior distribution lies above the FDA threshold. The key difference between the two methods is that the frequentist intervals reflect only the data, while the Bayesian intervals are also shaped by the prior. As Senn (2021) notes, the choice of prior in analyses like this one can be debated, since it does not necessarily reflect genuine prior belief. That said, both methods agree that vaccine exceeds the 30% FDA threshold.

A limitation of this analysis is that the Bayesian results are sensitive to the prior choice. Different prior would shift the posterior estimate. On the frequentist side, the close agreement between the large sample and bootstrap intervals suggests the asymptotic approximation works well.

In summary, both frameworks strongly support the conclusion that the BNT162b2 vaccine is highly effective, with the likelihood estimate near 95% and the Bayesian estimate near 75% under a pessimistic prior. The agreement across two different statistical frameworks makes the strong evidence for the vaccine's efficacy.

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Appendix

Code

```
# Plots for introduction
trial_matrix <- matrix(c(8, 17411-8, 162, 17511-162),
                      nrow = 2,
                      dimnames = list(Outcome = c("Infected", "Healthy"),
                                       Group = c("BNT162b2", "Placebo")))
rates <- trial_matrix["Infected", ] / colSums(trial_matrix)

barplot(rates,
        main="Figure 1: Infection Rate by Group",
        ylab="Proportion of Cases",
        col=c("skyblue", "salmon"),
        ylim=c(0, 0.01))

mosaicplot(as.array(trial_matrix),
           main="Figure 2: Mosaic Plot of Distribution",
           color = c("skyblue","salmon"),
           cex.axis = 0.66)
```

```
# Codes used for MLE Inference
```

```
# Data
```

```

t_obs <- 8
n_total <- 170
pi_hat <- t_obs / n_total

# MLE of psi
psi_hat <- (n_total - 2*t_obs) / (n_total - t_obs)

# Log-likelihood function in terms of psi
loglik.psi <- function(psi, t, n){
  if(psi >= 1)
    return(NA)
  t * log(1 - psi) - n * log(2 - psi)
}

# Fisher information and SE at psi_hat
fisher_info <- t_obs/(1 - psi_hat)^2 + n_total/(2 - psi_hat)^2
se <- 1/sqrt(fisher_info)

# Large sample CI
L <- psi_hat - (qnorm(0.975) * se)
U <- psi_hat + (qnorm(0.975) * se)

# Bootstrap CI
set.seed(123)
B <- 10000
boot_func <- function(i){
  t_star <- rbinom(1, size = n_total, prob = pi_hat)
  psi_star <- (n_total - 2*t_star) / (n_total - t_star)
  return(psi_star)
}

```

```

}

boot_psi <- c(unlist(lapply(1:B, boot_func)))
ci_boot <- quantile(boot_psi, c(0.025, 0.975))

# LRT
psi0 <- 0.30
pi0 <- (1 - psi0) / (2 - psi0)
W_obs <- 2 * (loglik.psi(psi_hat, t_obs, n_total) -
              loglik.psi(psi0, t_obs, n_total))
pval_chisq <- pchisq(W_obs, df = 1, lower.tail = FALSE)

# Empirical p-value
set.seed(123)
sim_func <- function(i){
  t_sim <- rbinom(1, size = n_total, prob = pi0)
  if(t_sim == 0 | t_sim == n_total)
    return(0)
  psi_sim <- (n_total - 2*t_sim) / (n_total - t_sim)
  w_star <- 2 * (loglik.psi(psi_sim, t_sim, n_total) -
                loglik.psi(psi0, t_sim, n_total))
  return(w_star)
}

Wstar <- c(unlist(lapply(1:B, sim_func)))
emp_p_val <- sum(Wstar >= W_obs) / B

# Taylor Approximation plot

# psi scale
loglik.psi.scalar <- function(psi){
  if(psi >= 1)

```

```

    return(NA)

    t_obs * log(1 - psi) - n_total * log(2 - psi)
}

ml_psi <- maxLik2(loglik = loglik.psi.scalar, start = 0.90)
plot(ml_psi) +
  labs(title = "Figure3: Second Order Taylor Approximation",
        x = expression(psi))

# pi scale
loglik_binom <- function(pi, obs_t, obs_n){
  if(pi <= 0 | pi >= 1)
    return(NA)

  obs_t * log(pi) + (obs_n - obs_t) * log(1 - pi)
}

ml_pi <- maxLik2(loglik = loglik_binom, start = 0.5,
                 obs_t = 8, obs_n = 170)
plot(ml_pi) +
  labs(title = "Figure4: Second Order Taylor Approximation",
        x = expression(pi))

```

Codes used for Bayesian Inference

Prior Elicitation

We want $P(\psi \geq 0.3) = 0.05$.

Using the transformation $\pi = (1-\psi)/(2-\psi)$:

If $\psi = 0.3$, then $\pi = (1-0.3)/(2-0.3) = 0.7/1.7 = 0.41176$

We look for a Beta(a, b) where the 5th percentile is 0.41176 and the median is 0.5.

```

prior_params <- beta.select(quantile1 = list(p = 0.05, x = 0.7/1.7),
                           quantile2 = list(p = 0.5, x = 0.5))

```

```

a_prior <- prior_params[1]

```

```

b_prior <- prior_params[2]

# Observed Data
t_obs <- 8
n_total <- 170

# Posterior via conjugacy: Beta(a + t, b + n - t)
a_post <- a_prior + t_obs
b_post <- b_prior + (n_total - t_obs)

# Posterior summaries on pi scale
post_median_pi <- qbeta(0.5, a_post, b_post)
post_hdi_pi <- hdi(qbeta, credMass = 0.95, shape1 = a_post, shape2 = b_post)

# Transform back to psi scale
post_median_psi <- (1 - 2 * post_median_pi) / (1 - post_median_pi)
post_hdi_psi <- (1 - 2 * post_hdi_pi) / (1 - post_hdi_pi)

# Posterior probability  $P(\psi > 0.3 \mid \text{data}) = P(\pi < 0.41176 \mid \text{data})$ 
prob_efficacy_gt_30 <- pbeta(0.7/1.7, a_post, b_post)

# Equal-tail CI on pi scale
ci_equal_pi <- qbeta(c(0.025, 0.975), a_post, b_post)
equal_L <- (1 - 2*ci_equal_pi[2]) / (1 - ci_equal_pi[2])
equal_U <- (1 - 2*ci_equal_pi[1]) / (1 - ci_equal_pi[1])

# Prior Posterior plot

x <- seq(0, 1, length.out = 500)
df <- data.frame(

```

```

pi = rep(x, 2),
pdf = c(dbeta(x, a_prior, b_prior), dbeta(x, a_post, b_post)),
distribution = rep(c("Prior", "Posterior"), each = 500)
)

ggplot(df, aes(x = pi, y = pdf, color = distribution)) +
  geom_line(linewidth = 1) +
  labs(title = "Figure5: Prior and Posterior Distribution",
       x = expression(pi),
       y = "Density") +
  theme_minimal() +
  scale_color_manual(values = c("Prior" = "blue", "Posterior" = "red"))

```

Proofs

The binomial distribution for T can be justified by the following argument. Since the sample sizes - n_1 and n_2 - in each group are large and the event rates are small, we may assume

$$X \approx \text{Pois}(n_1\pi_v) \quad \text{independently} \quad Y \approx \text{Pois}(n_2\pi_p)$$

This is because the Poisson distribution approximates a binomial well in this setting. Therefore the PMF of T can be found as following:

$$\begin{aligned}
P(T = t) &= P(X = t | X + Y = n) \\
&= \frac{P(X = t) \cdot P(Y = n - t)}{P(X + Y = n)} \\
&= \frac{\frac{e^{-n_1\pi_v}(n_1\pi_v)^t}{t!} \cdot \frac{e^{-n_2\pi_p}(n_2\pi_p)^{n-t}}{(n-t)!}}{\frac{e^{-(n_1\pi_v+n_2\pi_p)}(n_1\pi_v+n_2\pi_p)^n}{n!}} \\
&= \frac{n!}{t!(n-t)!} \cdot \frac{e^{-(n_1\pi_v)}(n_1\pi_v)^t \cdot e^{-(n_2\pi_p)}(n_2\pi_p)^{n-t}}{e^{-(n_1\pi_v+n_2\pi_p)}(n_1\pi_v+n_2\pi_p)^n} \\
&= \frac{n!}{t!(n-t)!} \cdot \frac{e^{-(n_1\pi_v+n_2\pi_p)} \cdot (n_1\pi_v)^t (n_2\pi_p)^{n-t}}{e^{-(n_1\pi_v+n_2\pi_p)} \cdot (n_1\pi_v+n_2\pi_p)^n} \\
&= \frac{n!}{t!(n-t)!} \cdot \frac{(n_1\pi_v)^t \cdot (n_2\pi_p)^{n-t}}{(n_1\pi_v+n_2\pi_p)^n} \\
&= \binom{n}{t} \cdot \frac{(n_1\pi_v)^t \cdot (n_2\pi_p)^{n-t}}{(n_1\pi_v+n_2\pi_p)^n} \\
&= \binom{n}{t} \cdot \frac{(n_1\pi_v)^t \cdot (n_2\pi_p)^{n-t}}{(n_1\pi_v+n_2\pi_p)^t \cdot (n_1\pi_v+n_2\pi_p)^{n-t}} \\
&= \binom{n}{t} \cdot \left(\frac{n_1\pi_v}{n_1\pi_v+n_2\pi_p} \right)^t \cdot \left(\frac{n_2\pi_p}{n_1\pi_v+n_2\pi_p} \right)^{n-t}
\end{aligned}$$

where $t = 0, 1, \dots, n$.

We know that $\pi = \frac{n_1\pi_v}{n_1\pi_v+n_2\pi_p} \approx \frac{\pi_v}{\pi_v+\pi_p}$ and $1 - \pi = \frac{n_2\pi_p}{n_1\pi_v+n_2\pi_p} \approx \frac{\pi_p}{\pi_v+\pi_p}$. Then,

$$P(T = t) = \binom{n}{t} \cdot \left(\frac{n_1\pi_v}{n_1\pi_v+n_2\pi_p} \right)^t \cdot \left(\frac{n_2\pi_p}{n_1\pi_v+n_2\pi_p} \right)^{n-t} = \binom{n}{t} \cdot \pi^t \cdot (1 - \pi)^{n-t}$$

Hence, $T \sim \text{Binom}(n, \pi)$, as desired.